

Project Proposal

Smart Hostel Management System

1. Introduction

Managing a university hostel manually can be time-consuming and inefficient. Students often face problems related to room allocation, complaint submission, and payment tracking. On the other hand, administrators find it difficult to manage large amounts of data using traditional methods.

To solve these issues, this project proposes a **Smart Hostel Management System**, a web-based application that will automate and simplify hostel management activities.

2. Objectives

The main objectives of this project are:

- To digitize the hostel management process
- To provide an easy system for room allocation
- To allow students to submit complaints online
- To maintain payment records efficiently
- To reduce manual workload for administrators

3. Scope of the Project

This system will be used within a university hostel environment. It will support multiple users with different roles such as students and administrators. The system will provide separate interfaces and functionalities based on user roles.

4. System Users

The system will have two types of users:

a) Admin:

- Manage student information

- Approve or reject room requests
- Handle and resolve complaints
- Monitor payment records

b) Student:

- Register and log in to the system
- Request hostel rooms
- Submit complaints
- View room and payment status

5. Features of the System

- **User Authentication System** (Login & Registration)
- **Room Allocation System** (Request and Approval)
- **Complaint Management System**
- **Payment Tracking System**
- **Dashboard for both Admin and Students**
- **Navigation Bar for easy page access**

6. Technology Stack

The project will be developed using the following technologies:

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Web Server (Node.js / similar)
- **Data Format:** JSON
- **API:** REST API for communication
- **AJAX:** For asynchronous data transfer without page reload

7. System Design Overview

The system will follow a client-server architecture:

- The **frontend** will handle user interaction and interface
- The **backend** will process requests and manage data
- **JSON files** will be used to store and retrieve data
- **AJAX** will enable smooth communication between frontend and backend

8. Expected Outcome

After completing the project, the system will:

- Reduce manual errors in hostel management
- Provide faster service to students
- Improve overall efficiency of administration
- Ensure better data organization and accessibility

9. Conclusion

The Smart Hostel Management System aims to modernize traditional hostel operations by introducing a digital platform. It will enhance both user experience and administrative efficiency. This project can be further extended in the future with advanced features like real-time notifications and database integration.

Proposed By,
Kuldip Saha Mugdha
IT22018
3rd year 2nd semester